Iftikhar Ahmed

Research Assistant, Big-Matrix R&D Lab

Seeking Master's and PhD positions

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Research interests: audio deepfake detection, adversarial robustness, cross-lingual speech processing, medical image analysis

Research Experience

Research Assistant Present

Big-Matrix R&D Lab · Team Leader: Prof. Mohammad Monir Uddin

- Contribute to model training workflows supporting the lab's research projects in AI for healthcare.
- Assist in academic paper writing and manuscript preparation.
- Perform data collection, annotation and segmentation for research datasets.

Publication

Iftikhar Ahmed, Mueid Islam Arian, Nusrat Jahan Lisa. "Mel-Spectrogram Representations for CNN Based Adversarial Defense for Audio Deepfake Detection." In *Proceedings of the 11th International Conference on Machine Learning Technologies (ICMLT 2026)*, Germany. IEEE, 2026.

IEEE Xplore DOI: 10.1109/ICMLT69916.2026.11689539

- Designed and evaluated a CNN-based (Xception) adversarial defense for audio deepfake detection. Mel-spectrograms reach an EER of 0.034 against 0.053 for LFCC, a 36% relative improvement, across seven neural vocoders on WaveFake.
- Implemented adaptive FGSM/PGD adversarial training, reducing EER under FGSM attack from 0.70 to 0.107 and under PGD attack from 0.90 to 0.154.

Education

North South University Dhaka, Bangladesh

B.Sc. in Computer Science and Engineering · CGPA 3.07/4.00 2021 – 2026

Undergraduate Thesis

LectureAssist: An Agentic Multimodal AI Assistant that Turns Lectures into Adaptive Study Material 2026

North South University · Supervisor: Dr. Md Adnan Arefeen · With Salman Shafi and Jarinah Tasnim

Scene-adaptive OCR and Whisper transcription feed a multi-agent pipeline that plans, generates, validates and refines five question types, served by Gemma 3 on a single consumer GPU, all locally hosted. Agentic and one-shot generation are compared with an LLM-as-judge evaluation.

Research Projects

Polyp Segmentation in Colonoscopy Images Code

Biomedical image segmentation

U-Net with an ImageNet-pretrained ResNet34 encoder on Kvasir-SEG, trained with BCE + Dice loss. Mean test Dice 0.910 and IoU 0.855 at original resolution, with a failure analysis on pedunculated and flat polyps.

Emotion Recognition from Hand-Sign Gestures

Computer vision

A computer vision system that identifies a user's emotion from hand-sign gestures.

Safety Equipment Detection Code

Object detection

A YOLOv5 object detector for safety equipment.

Technical Skills

Machine learning: PyTorch, scikit-learn, NumPy, Hugging Face, CNNs (Xception, U-Net, ResNet), YOLOv5, Whisper, adversarial training (FGSM, PGD)

Languages & tools: Python, LaTeX, Git, Kaggle and Colab GPU workflows

References

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